

Platform Layers & Growth Roadmap

A reading brief — current capabilities and the direction ahead

EnergyLens is an energy intelligence and decision-support platform for buildings, built on Microsoft Fabric. It is not a dashboard: it answers four questions — **what is happening, why, what should be done, and what would happen if we changed something**. This brief summarises what the platform does today and the direction I am developing it in next.

1 · The platform today

The core engine is built and runs end-to-end on a medallion data architecture, with a customer-facing web application on top.

Layer	What it does
Data platform	Microsoft Fabric Lakehouse — medallion (Bronze → Silver → Gold), Delta Lake
KPI engine	Energy Use Intensity, peak demand, solar, battery, heat-pump COP, carbon intensity
Detection	Rule-based anomaly detection + automated fault detection & diagnostics (HVAC)
Simulation	Physics-based what-if: solar, battery, heat-pump, insulation, windows
Sustainability	CO ₂ accounting, ESG reporting, EU Taxonomy alignment
Compliance	MEPS renovation radar, CRREM stranding, ESRS-aligned reporting support
Web app	Portfolio, actions, alerts, solar, compliance, AI copilot — multi-tenant, billing
Current focus	Commercial buildings: office, retail, hotel, healthcare, education, logistics

2 · Where I am taking it

A common critique is that energy management looks unattractive today given energy prices and the cost of smart-building hardware. My conclusion is that this is a **positioning issue, not a product flaw**: EnergyLens is a software layer where hardware is optional, and the right unit of sale is a **portfolio**, not a single building.

Two segments

- **Commercial** (today) and **Residential** (new) — the same engine, two markets.

Residential pivot

- The buyer is the **property manager / housing company**, not the individual flat. Energy use is heating-dominated and the decision is regulation- and capital-driven: *which buildings to renovate first, which measures, what payback*.

A resident layer

- Each tenant sees their own consumption, their share of common-area energy, and an anonymised comparison to the building average.

Why now — a regulatory tailwind.

From **1 January 2027**, German law (EU Energy Efficiency Directive, via the Heizkostenverordnung) requires landlords to give tenants **monthly** consumption information; if they fail, a tenant may withhold **3%** of their heating costs. EnergyLens turns this legal obligation into a product feature — managers comply automatically, and the resident view becomes sticky.

Two cross-cutting differentiators

- **Forward-cost ROI + abatement curve** — price the *rising* CO₂ / tariff cost into every payback, so the cost of doing nothing visibly grows; and rank every measure across the portfolio by cost per tonne avoided.
- **Hardware-free onboarding** — start from utility bills and existing sub-metering, with no new sensors. This directly answers the cost objection.

Market focus

- **Germany first** (strongest regulation, subsidies and carbon price), **Turkey second** (rising tariffs and the urban-renewal wedge).

3 · Architecture & roadmap

These directions **extend the existing platform** rather than replacing it. The work is organised into four architecture blocks, delivered in dependency order.

- **User & access** — new Residential building type, unit-level data, resident accounts.
- **Data connectivity** — hardware-free ingestion from bills and sub-metering.
- **Packaging & payment** — segment-based plans; residents included for free.
- **App experience** — segment-aware navigation plus a lightweight resident view.

Phase	Focus
P1	Data model + access control (the foundation)
P2	Hardware-free data ingestion (bills → sub-metering)
P3	Manager-facing residential analytics
P4	Resident view + commercial packaging

Why this is differentiated

- **Compliance is the wedge** — durable, and independent of today's energy price.
- **The cost of doing nothing rises** with carbon prices and tariffs.
- **No hardware required** — a low barrier to adopt.
- **A conversational AI copilot** explains the 'why' and drafts the plan.

In short: the same proven engine, repositioned for the buyers and the regulation that make it compelling now.

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